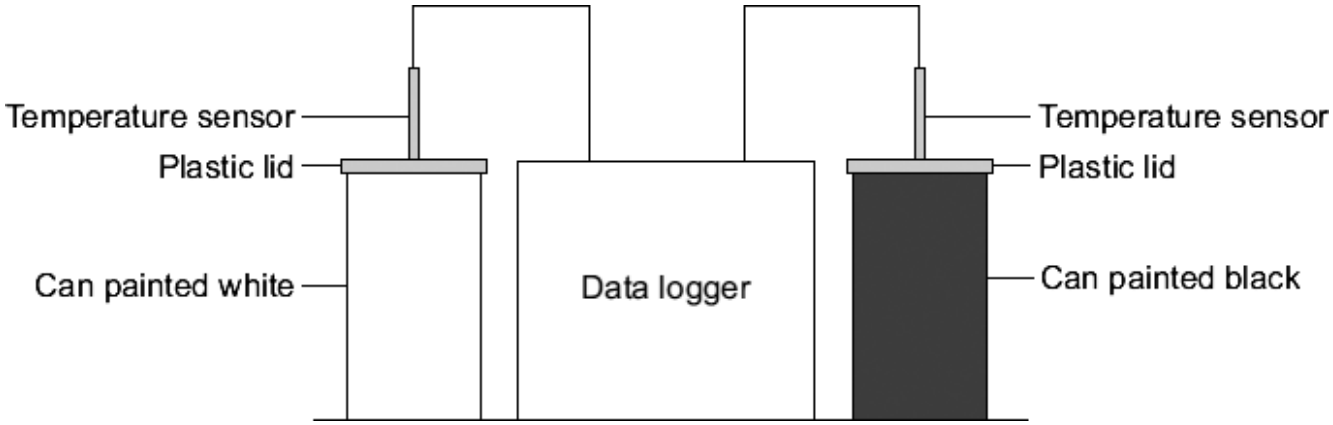
Q1.

A student investigated the emission of infrared radiation from two cans.

- The two cans were the same size.
- One can was painted white and the other can was painted black.
- The student poured the same volume of boiling water into each can.
- A data logger recorded the temperature in each can for the next 14 minutes.



(a) Draw a ring around the correct answer in the box to complete the sentence.

The plastic lids prevent energy loss by

conduction.
convection.
infrared radiation.

(1)

(b) The data logger takes two temperature readings each second, and then plots a graph for the student.

The student could have used a thermometer to measure the temperature.



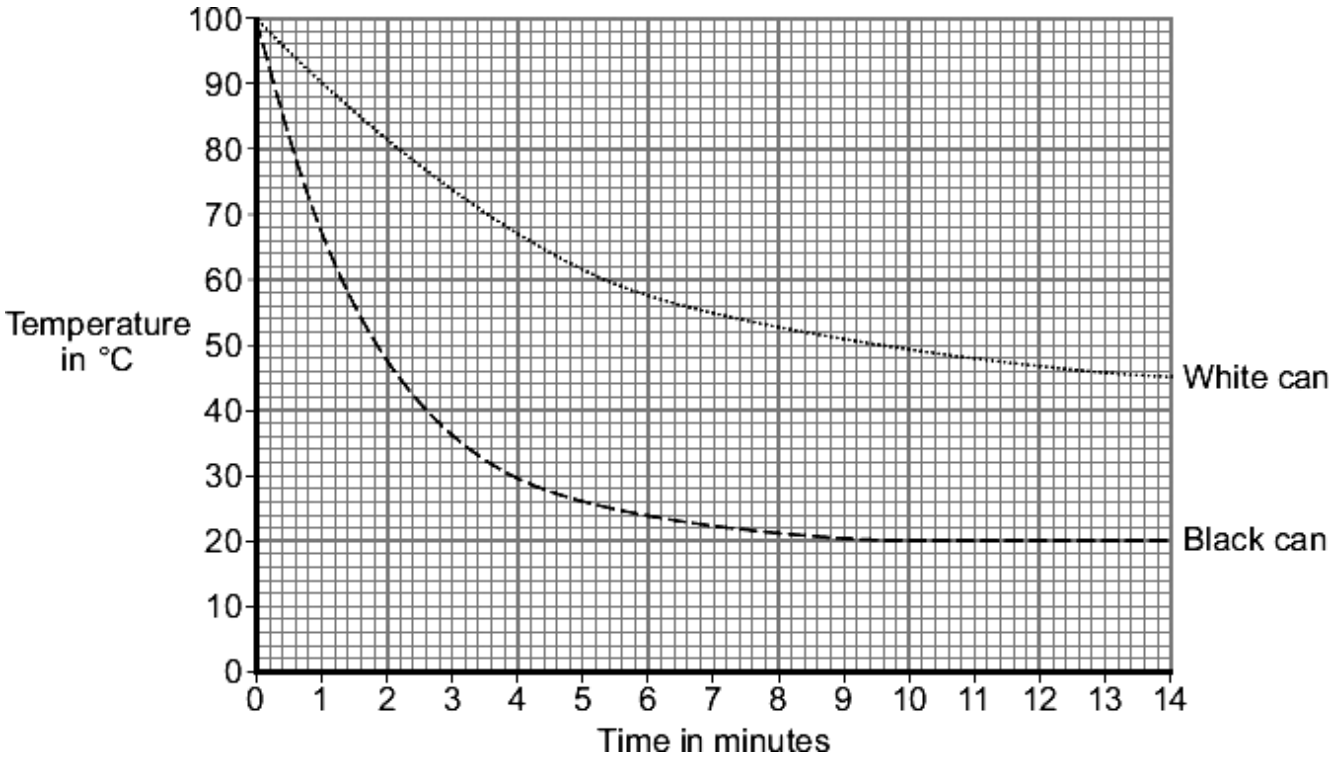
Give **two advantages** of using a data logger instead of the thermometer

1. _____

2. _____

(2)

(c) The graph shows the student's results.



(i) Explain the difference in temperature between the water in the two cans after 14 minutes.

_____

_____

_____

_____

(2)

(ii) The student decided to repeat the investigation using a can painted grey.

On the graph, draw a line for the results you would expect for the can painted grey.

(2)

(Total 7 marks)

Q1.

(a) convection

1

(b) any **two** from:

- easier / clearer to read **or** reading error less likely / takes readings for you
- greater resolution / temperature displayed to two decimal places
  ignore precision / accurate / reliable / sensitivity
- more frequent readings
  ignore readings every second / plots graph / quicker
- more data
- live graphical representation
  allow references to safety
  allow no need to open lid (thus preventing cooling)

2

(c) (i) black is a better <u>emitter</u> / <u>radiator</u> of energy / infrared radiation
allow heat
for full marks there must be a comparison in terms of emission / radiation
*allow **1** mark for: temperature of black can decreases faster / more*
or
water in black can has reached room / constant temperature whilst water in white can is still cooling
or
correct description of temperature drop in both cans after 14 minutes

2

(ii) (any) line drawn between the white can and black can lines on graph **starting at 100** (°C)

1

line shows the same trend as white can line **or** black can line
second mark only scores if first marking point correct
*do **not** award this mark if line shows intentional rise at any point*

1

[7]